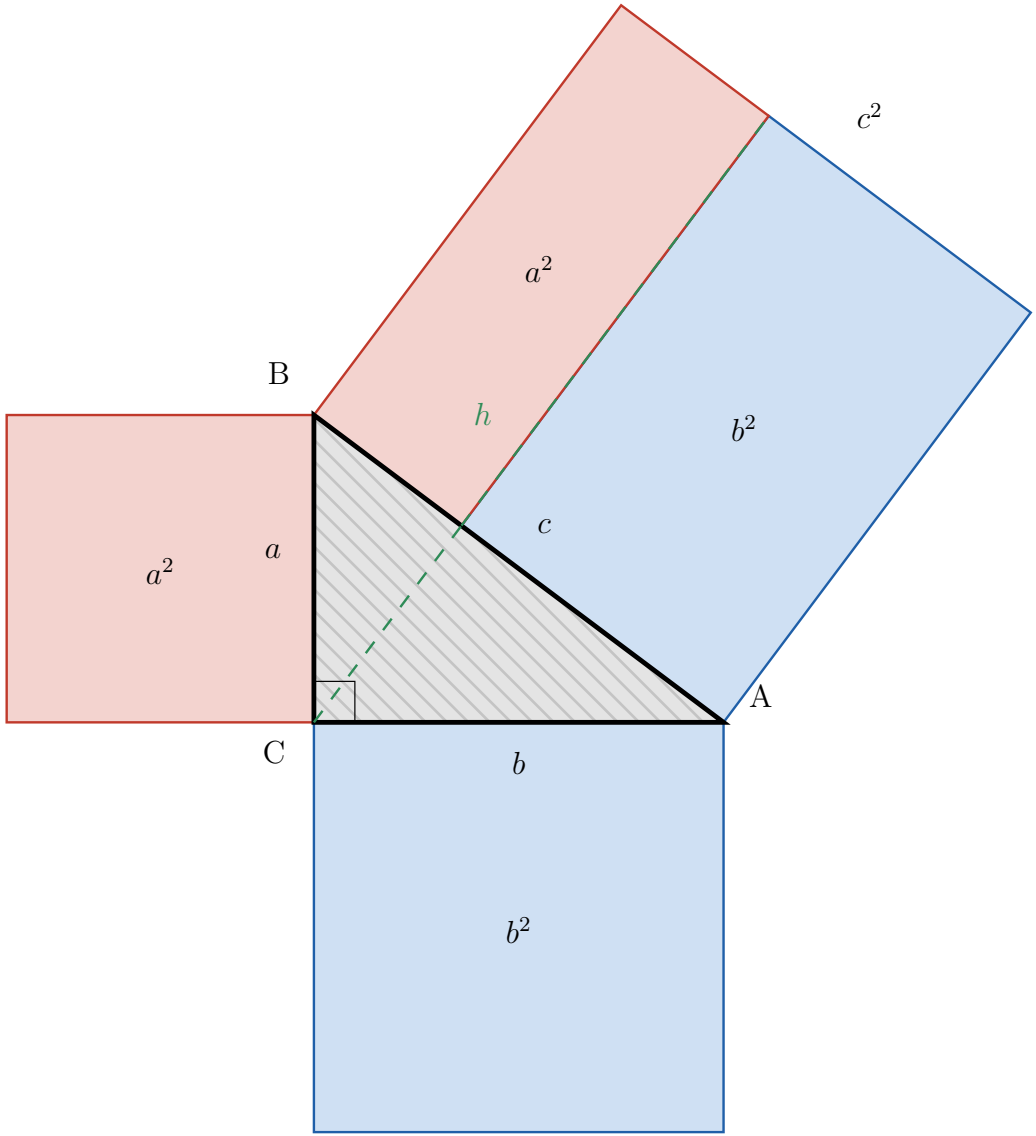
Pythagorean theorem

Euclid's figure: the squares on the legs equal the square on the hypotenuse



$$a^2 + b^2 = c^2$$

Areas for $a = 3$, $b = 4$, $c = 5$:

- $a^2 = 9$
- $b^2 = 16$
- $c^2 = 25$

$9 + 16 = 25$

The altitude h from the right angle cuts the big square in two rectangles: blue $= b^2$, red $= a^2$.